



ULAB

UNIVERSITY OF LIBERAL ARTS
BANGLADESH

Final Assignment

Comprehensive Class Notes

A complete, date-wise compilation (Parts 1–2)

Course: NSC4182 — Mathematics

Section: 1

Semester: Summer 2026

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Contents

Part 1 (May 21 – Jun 25): Sets, Number Systems, Absolute Value, Polynomials, Exponents, Factoring, Functions, Equations, Graphing, Circles

Part 2 (Jul 11 – Aug 20): Quadratics, Geometry, Coordinate & Polar Systems, Functions & Graphs, Intro to Derivatives, Linear Algebra, Cramer's Rule, Exponential Growth, Graph Analysis

Colour Legend: **Blue** = method/concept **Teal** = formula/definition **Orange** =
key value **Green** = final answer **Red** = corrected error

Abstract

This document is a complete, cleaned-up compilation of the class notes for **NSC4182 — Mathematics (Section 1, Summer 2026)**, taught by Md. Al-Amin. It reorganises a full semester of handwritten lecture notes into a single, typeset reference, arranged in two date-wise parts split at 11 July: **Part 1** gathers the foundational material (sets, number systems, absolute value, polynomials, exponents, factoring, functions, equations, and circles), while **Part 2** covers the advanced topics (quadratics, geometry and coordinate systems, functions and their graphical behaviour, an introduction to derivatives, linear algebra with Cramer's rule, exponential growth, and graph analysis).

Every in-scope worked example from the lectures has been transcribed and verified. Incomplete problems have been carried through to a final answer, and genuine errors in the original notes have been corrected and flagged. Throughout, a consistent colour scheme distinguishes methods, formulas, key values, final answers, and corrections, so the notes double as both a revision resource and an exam-preparation aid. The mathematical conventions of the course — in particular the instructor's strict first-row cofactor expansion for determinants — are preserved.

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Part 1 — Foundations (May 21 – June 25)

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Class 02	May 23	Sets (continued)
Class 03	June 03	Number hierarchy, expressions vs. equations
Class 04	June 04	Polynomials, exponents, scientific notation, applied problems
Class 05	June 11	Factoring, completing the square, grouping
Class 07	June 18	Solving equations, functions, domain & range
Class 08	June 20	Rational equations, graphing basic equations
Class 09	June 25	Circles (standard & general form)

Part 2 — Advanced Topics (July 11 – August 20)

Class	Date	Topic
Class 10	July 11	Quadratic equations (four methods, discriminant, complex roots)
Class 11	July 16	Triangles & Pythagorean applications
Class 12	July 18	Applied quadratics (screens, pool border, baseball)
Class 13	mid-July	Coordinate systems & polar coordinates
Class 14	mid-July	Distance, midpoint, slope & triangles
Class 15	July 25	Functions, domain & range, vertical-line test
Class 16	August 01	Even & odd functions
Class 17	August 06	Slopes, increasing/decreasing & local extrema
Class 18	August 08	Average rate of change, secant lines, intro to derivatives
Class 19	August 13	Linear algebra, Gauss-Jordan, Cramer's rule, determinants
Class 20	August 15	Exponential growth model
Class 21	August 20	Lines, parabolas & graph analysis

Learning Outcomes

What I learned across the semester

Foundations & Number Sense

- I can define a set and express it in descriptive, set-builder, and roster form, and carry out union, intersection, and complement operations against a universal set.
- I understand the hierarchy of number systems ($\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$) and can classify a given number correctly.
- I can apply the properties of real numbers (commutative, associative, distributive, identities, inverses) and evaluate expressions using the correct order of operations.
- I learned to read the absolute-value rule correctly ($|a| = -a$ when $a < 0$, not -1) and to solve absolute-value equations and inequalities, including converting to interval notation.

Algebra & Polynomials

- I can add, subtract, multiply, and divide polynomials, identify the degree and leading coefficient, and distinguish polynomials from non-polynomials.
- I can factor using common techniques — difference of squares, sum/difference of cubes, perfect-square trinomials, splitting the middle term, completing the square, and grouping.
- I learned the laws of exponents (including the power-of-a-power rule $(a^m)^n = a^{mn}$) and scientific notation, and can apply them to simplify and to model real quantities.

Equations, Functions & Graphs

- I can solve linear, rational, and quadratic equations, choosing among factoring, the square-root method, completing the square, and the quadratic formula, and I can interpret the discriminant $b^2 - 4ac$ to predict the nature of the roots.
- I understand what makes a relation a function, can find the domain and range (recognising that only the denominator restricts the domain), and can apply the vertical-line test.
- I can determine whether a function is even, odd, or neither, both algebraically and from graphical symmetry, and read absolute and local maxima/minima from a graph.
- I can analyse a parabola completely — intercepts, vertex via $x = -b/2a$, domain and range — and find the intersection points of a line and a curve without stopping at the first root.

Geometry, Coordinate Systems & Calculus Foundations

- I can apply the Pythagorean theorem and the distance, midpoint, and slope formulas, test a triangle for a right angle, and compute its area (by base–height and by determinant).
- I can convert points and equations between Cartesian and polar coordinate systems and identify the four coordinate systems (Cartesian, polar, cylindrical, spherical).
- I understand the average rate of change as the slope of a secant line, and I was introduced to the derivative and the power rule for finding critical points.

Linear Algebra & Growth Models

- I can distinguish linear from non-linear equations, write a system in matrix form $Ax = B$, and build the augmented matrix.
- I can solve a system by Gauss-Jordan elimination and by Cramer's rule, and I can evaluate determinants using strict first-row cofactor expansion.
- I can derive and apply the exponential growth model $N(t) = N_0e^{kt}$: finding the growth rate k , projecting future values, and solving for the time to reach a target (dividing the exponent appropriately when the data point is not at $t = 1$).

Part 1

Foundations (before 11 July)

May 21 – June 25

What this part covers

Sets & Set Operations • Number Systems • Properties of Real Numbers • Number Line & Distance • Approximating Decimals • Absolute Value Equations & Inequalities • Scientific Notation • Order of Operations • Polynomials (add, multiply, divide) • Laws of Exponents • Factoring • Solving Equations • Functions, Domain & Range • Rational Equations • Graphing Basic Equations • Circles (standard & general form)

Class 01

Date: May 21

Topic: Introduction & Sets

What is Mathematics?

Mathematics is a tool for thinking logically and solving problems systematically. We use maths in daily life in almost every part of it, knowingly or unknowingly.

Course Grading Breakdown

Mid	25%	Assignments	10 + 5%
Final	40%	Quizzes	15%
Attendance	5%		

Bonus marks: +5 (bonus) and +2 (full attendance)

Sets

A **set** is a well-defined collection of distinct objects. The objects are called **elements**, and are expressed inside curly braces.

$$A = \{-5, -3, -1, 1, 3, 5\} \quad B = \{2, 3, 5, 7, 11, 13\}$$

Three forms to express a set:

1. **Descriptive form** — e.g. “The set of all vowels in the English alphabet”
2. **Set-builder form** — e.g. $\{x : x \text{ is a vowel in the English alphabet}\}$
3. **Roster / Tabular form** — e.g. $\{a, e, i, o, u\}$

Set Operations

Example — Given $A = \{1, 2, 3, 5, 8\}$, $B = \{3, 5, 7\}$, $C = \{2, 4, 6, 8\}$:

(a) **Intersection** $A \cap B$ (elements common to both)

$$A \cap B = \{1, 2, 3, 5, 8\} \cap \{3, 5, 7\} = \{3, 5\}$$

(b) **Union** $A \cup B$ (all elements from both)

$$A \cup B = \{1, 2, 3, 5, 7, 8\}$$

(c) $B \cap (A \cup C)$

$$A \cup C = \{1, 2, 3, 4, 5, 6, 8\}$$

$$B \cap (A \cup C) = \{3, 5, 7\} \cap \{1, 2, 3, 4, 5, 6, 8\} = \{3, 5\}$$

Complement

If A is a set, the **complement** of A , denoted $\overline{A}$, is the set of all elements in the universal set U that are *not* in A .

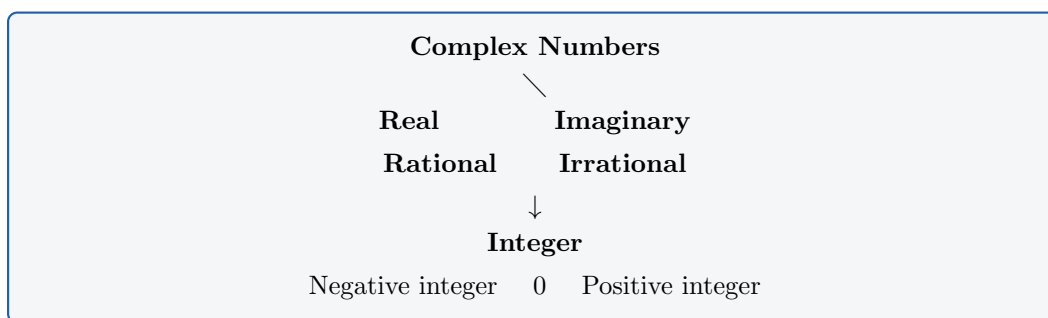
$$\overline{A} = U - A \quad (\text{everything} - A)$$

Example: $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{1, 3, 5, 7, 9\}$

$$\overline{A} = \{2, 4, 6, 8\}$$

Number Systems

Every complex number has the form $z = a + ib$, where a is the **real** part and b is the **imaginary** part.



The set of real numbers is a subset of the complex numbers.

Properties of Real Numbers

- **Addition** is **commutative** (order), **associative** (grouping)
- 0 is the **additive identity**; $-a$ is the negative (additive inverse) of a
- **Multiplication:** 1 is the **multiplicative identity**: $a \cdot 1 = a$
- Multiplying a non-zero number by its **reciprocal** yields 1: $a \cdot \frac{1}{a} = 1$
- Multiplication is **distributive** over addition:

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc$$

Rational Numbers

Numbers that can be written as a fraction $\frac{p}{q}$, $q \neq 0$.

Common Number Sets

$\mathbb{N}$	Natural numbers	$\mathbb{Q}$	Rational numbers
$\mathbb{W}/\mathbb{N}_0$	Whole numbers	$\mathbb{I}$	Irrational numbers
$\mathbb{Z}$	Integers	$\mathbb{R}$	Real numbers
$\mathbb{C}$	Complex numbers		where $i = \sqrt{-1}$

Example — classify $\{-3, \frac{4}{3}, 0.12, \sqrt{2}, \pi, 10, 2.1515\dots\}$:

- | | |
|----------------|--|
| (a) Natural | 10 |
| (b) Integers | $-3, 10$ |
| (c) Rational | $-3, \frac{4}{3}, 10, 0.12, 2.1515\dots$ |
| (d) Irrational | $\sqrt{2}, \pi$ |
| (e) All | are real numbers |

Approximating a Decimal to Two Places

Two methods: **(a) Truncating** and **(b) Rounding**. For rounding, if the next digit $x \geq 5$, round up.

Example: $20.98\bar{7} \dots \rightarrow$ ceiling value 21, floor value 20; rounding $20.98\bar{7} = \mathbf{20.99}$

Round to two decimals:

- | | | |
|---------------|---------------|--------|
| (a) 3.14159 | $\rightarrow$ | 3.14 |
| (b) 0.056128 | $\rightarrow$ | 0.06 |
| (c) 893.46125 | $\rightarrow$ | 893.46 |

Distance on the Number Line

Distance between two points = $d(P, Q) = |Q - P|$ (always positive).

Given $A = -5$, $B = -3$, $C = 1$, $D = 6$, $O = 0$:

$$\begin{aligned} d(A, B) &= |-3 - (-5)| = |-3 + 5| = |2| = \mathbf{2} \\ d(C, B) &= |-3 - 1| = |-4| = \mathbf{4} \\ d(O, A) &= |-5 - 0| = |-5| = \mathbf{5} \\ d(C, D) &= |6 - 1| = |5| = \mathbf{5} \end{aligned}$$

Absolute Value

- If $a \geq 0$, then $|a| = a$
- If $a < 0$, then $|a| = -a$

Correction: the notes wrote “if $a < 0$, then $|a| = -1$ ”. The rule is $|a| = -a$ (negate the value), not -1 . E.g. $|-7| = -(-7) = 7$.

Example — solve absolute-value equations:

(a) $|x + 4| = 13$

$$x + 4 = 13 \Rightarrow x = 9 \quad \text{or} \quad x + 4 = -13 \Rightarrow x = -17$$

Solution set: $\{-17, 9\}$

(b) $|2x - 3| + 2 = 7 \Rightarrow |2x - 3| = 5$

$$2x - 3 = 5 \Rightarrow 2x = 8 \Rightarrow x = 4 \quad \text{or} \quad 2x - 3 = -5 \Rightarrow 2x = -2 \Rightarrow x = -1$$

Solution set: $\{-1, 4\}$

Absolute Value Inequality

Example — solve $|2x + 4| \leq 3$:

$$\begin{aligned} -3 &\leq 2x + 4 \leq 3 \\ -3 - 4 &\leq 2x \leq 3 - 4 \\ -7 &\leq 2x \leq -1 \\ -\frac{7}{2} &\leq x \leq -\frac{1}{2} \end{aligned}$$

$$x \in \left[-\frac{7}{2}, -\frac{1}{2}\right]$$

Class 02

Date: May 23

Topic: Sets (continued) — see Class 01

Class 02 reinforced the set material from Class 01: set definition, the three forms of expressing a set, and the set operations (intersection, union, complement). The worked examples above cover this content.

Class 03

Date: June 03

Topic: Number Hierarchy & Expressions

Number Hierarchy (review)

Real numbers $\rightarrow$ Rational & Irrational; Rational $\rightarrow$ Integers $\rightarrow$ Whole $\rightarrow$ Natural. Note: **0 is not a natural number**. Complex/imaginary: $i = \sqrt{-1}$, e.g. $x^2 + 1 = 0$ has no real solution.

Expression vs. Equation

Expression	Equation
Numeric: $(1, 2, 3)$	$5 + 3 = 8$
Algebraic: (x, y)	Has two sides

Class 04

Date: June 04

Topic: Polynomials, Exponents & Scientific Notation

Monomials and Polynomials

A **polynomial** is “a sentence in maths.” Example: $3x^2 + 5x - 7$ has **3 terms**: $3x^2$, $5x$, -7 . General monomial term: ax^k , where $k \geq 0$ is an integer.

A **monomial** (e.g. $7x^2$, $-3x$, 9 , $\frac{1}{2}x^5$) satisfies:

1. Only one term
2. Variable power must be a whole number (no negatives)
3. No variable in the denominator: $\frac{5}{x}$ is **not** allowed, but $\frac{5}{2}x$ is fine

A **polynomial** satisfies:

1. Made of terms, e.g. $3x^2 + 5x - 1$
2. Powers of variables must be whole numbers (no negatives)
3. No variable in the denominator

Each term has a coefficient multiplied by a variable raised to a **non-negative integer** power (Ax^n).

Addition of Polynomials

$$(8x^3 - 2x^2 + 6x - 2) + (3x^4 - 2x^3 + x^2 + 3)$$

$$= 3x^4 + 6x^3 - x^2 + 6x + 1$$

Subtraction: $(8x^3 - 2x^2 + 6x - 2) - (3x^4 - 2x^3 + x^2 + 3)$

$$= -3x^4 + 10x^3 - 3x^2 + 6x - 5$$

Multiplication of Polynomials

$$(2x + 5)(x^2 - x + 2) = 2x(x^2 - x + 2) + 5(x^2 - x + 2)$$

$$= 2x^3 - 2x^2 + 4x + 5x^2 - 5x + 10 = 2x^3 + 3x^2 - x + 10$$

Division of Polynomials

Numeric analogy: Divide 842 by 15:

$$842 \div 15 = 56 \text{ remainder } 2$$

Degree of a polynomial = degree of the **leading term** (first term).

Long division: $\frac{x^4 - 3x^3 + 2x - 5}{x^2 - x + 1}$

Quotient: $x^2 - 2x - 3$, Remainder: $x - 2$

Order of Operations

$$3 + 4 \cdot 5 = 3 + 20 = 23$$

$$8 \cdot 2 + 1 = 16 + 1 = 17$$

$$(3 + 4) \cdot 5 = 7 \cdot 5 = 35$$

$$5 \times (3 + 4) + 2 = 5 \cdot 7 + 2 = 37$$

Scientific Notation

Form: $a = c \times 10^n$, where $1 \leq c < 10$ and n is an integer.

$$513 = 5.13 \times 10^2$$

$$0.0000000043 = 4.3 \times 10^{-10}$$

$$93,000,000 = 9.3 \times 10^7$$

$$7.3 = 7.3 \times 10^0$$

Applied Problems

(1) **Production cost.** The weekly cost of manufacturing x watches is $C = 4000 + 2x$ (dollars).

(a) $x = 1000$: $C = 4000 + 2(1000) = 4000 + 2000 = \text{\$6000}$

(b) $x = 2000$: $C = 4000 + 2(2000) = 4000 + 4000 = \text{\$8000}$

(2) **Astronomy — speed of light.** Speed = 186,000 miles/second.

$$v = \frac{s}{t} \quad \text{where } v = \text{velocity, } s = \text{distance (movement), } t = \text{time.} \quad \text{So } t = \frac{s}{v}.$$

Example: time for light to travel 93,000,000 miles:

$$t = \frac{93,000,000}{186,000} = 500 \text{ seconds} = 5 \times 10^2 \text{ s}$$

Laws of Exponents

$$a^m \cdot a^n = a^{m+n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

$$\frac{a^m}{a^n} = a^{m-n} = \frac{1}{a^{n-m}} \quad (a \neq 0)$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \quad (b \neq 0)$$

Note: the board line “ $(a^m)^n = a^{m+n}$ ” is a slip — the power-of-a-power rule is $(a^m)^n = a^{mn}$ (multiply exponents). Only $a^m \cdot a^n = a^{m+n}$ uses addition.

Example: $\frac{x^{-2}}{x^{-5}} = x^{-2-(-5)} = x^3$

Example: $\left(\frac{x^{-3}}{3y^{-1}}\right)^{-2}, \quad x \neq 0, y \neq 0$

$$= \frac{x^6}{3^{-2}y^2} = \frac{9x^6}{y^2}$$

Example — evaluate at $x = 3, y = -4$:

(c) $\frac{3y}{2-2x} = \frac{3(-1)}{2-2(3)} = \frac{-3}{-4} = \frac{3}{4}$

(d) $|-4x + y| = |-4(3) + (-4)| = |-12 - 4| = |-16| = 16$

Watch the substitution: in (c) the notes used $y = -1$; with the stated $y = -4$ recompute carefully. In (d) the notes reached 13 by using $y = -1$ ($|-12 - 1| = 13$); with $y = -4$ the answer is 16. Always substitute the value you were actually given.

Class 05

Date: June 11

Topic: Factoring

Polynomial Definition (formal)

$$ax^n + bx^{n-1} + \cdots + z, \quad a \neq 0$$

where a is the **leading coefficient**.

Identify which are polynomials: among $-\emptyset$, $x^{2/3}$, 3^x , 4 , $\frac{x^3}{2} + y$

4 (degree 0) and $\frac{x^3}{2} + y$ are polynomials. $x^{2/3}$ (fractional power) and 3^x (variable exponent) are **not**.

Factoring — Worked Examples

$$(1) \quad x^2 - 4 = (x - 2)(x + 2)$$

$$(3) \quad x^3 - 1 = (x - 1)(x^2 + x + 1)$$

$$(4) \quad x^3 + 8 = (x + 2)(x^2 - 2x + 4)$$

$$(5) \quad x^4 - 16 = (x^2 - 4)(x^2 + 4) = (x - 2)(x + 2)(x^2 + 4)$$

$$(6) \quad x^2 + 6x + 9 = (x + 3)^2 \quad [x^2 + 2 \cdot x \cdot 3 + 3^2]$$

$$(7) \quad 9x^2 - 6x + 1 = (3x - 1)^2$$

$$(8) \quad 25x^2 + 30x + 9 = (5x + 3)^2$$

$$(9) \quad x^2 + 7x + 10 = x^2 + 2x + 5x + 10 = (x + 2)(x + 5)$$

$$(10) \quad x^2 - 6x + 8 = x^2 - 2x - 4x + 8 = (x - 2)(x - 4)$$

$$(11) \quad x^2 - x - 12 = (x + 3)(x - 4)$$

$$(12) \quad x^2 + 4x - 12 = (x - 2)(x + 6)$$

Factoring a Trinomial (split the middle term)

$$2x^2 - x - 6 = (2x^2 - 4x) + (3x - 6) = 2x(x - 2) + 3(x - 2) = (x - 2)(2x + 3)$$

Completing the Square

To complete the square on $y^2 + 8y$: take $(\frac{1}{2} \cdot 8)^2 = 16$ and add it.

$$y^2 + 8y + 16 = (y + 4)^2$$

Factoring by Grouping

(14)

$$(x^2 + 2)x + (x^2 + 2) \cdot 3 = (x^2 + 2)(x + 3)$$

(15) $3(x - 1)^2(x + 2)^3 + 4(x - 1)^3(x + 2)^3$

$$= (x - 1)^2(x + 2)^3[3(x + 2) + 4(x - 1)]$$

$$= (x - 1)^2(x + 2)^3(3x + 6 + 4x - 4)$$

$$= (x - 1)^2(x + 2)^3(7x + 2)$$

$$3(x - 1)^2(x + 2)^3 + 4(x - 1)^3(x + 2)^3 = (x - 1)^2(x + 2)^3(7x + 2)$$

Class 07 (Day 07)

Date: June 18

Topic: Solving Equations & Functions

Solving a Linear Equation

Example: Solve $3x - 5 = 4$.

$$3x - 5 = 4$$

$$3x = 9$$

$$x = \frac{9}{3} = 3$$

$$x = 3$$

Types of Equations (by degree)

The **degree** (highest power of the variable) names the equation type:

Linear	$x^1 = 0$	$ax + b = 0$
Quadratic	$x^2 = 0$	$ax^2 + bx + c = 0$
Cubic	$x^3 = 0$	$ax^3 + bx^2 + cx + d = 0$

A quartic would be $x^4 + x^3 + x^2 + x^1 = 0$, and so on.

Functions — Two Conditions

A relation is a **function** if it satisfies two conditions:

1. Every entry of the domain must map to the co-domain.
2. One domain value cannot map to two different outputs.

Examples of functions: $y = x^2$, $y = \sqrt{x}$, $f(x) = x^3$.

Example — $y = x^3$ mapping $\{1, 2, 3\}$:

Domain	$\{1, 2, 3\}$
Range	$\{1, 8, 27\}$
Co-domain	$\{1, 8, 27, 64, 125\}$

Range = the outputs actually hit. **Co-domain** = the full target set (may be larger than the range).

Domain of a Rational Function

Example: $\frac{5}{x-2}$

Domain: $\{x \mid x \neq 2\}$ (denominator cannot be zero)

Only the denominator restricts the domain. The board wrote “ $x \neq \mathbb{Z}$ ” — the actual restriction is just $x \neq 2$ (the value that makes the denominator 0). Numerator zeros do *not* restrict the domain.

Domain & Range of a Polynomial

Example: $f(x) = 3x^3 - 5$

Domain: $(-\infty, +\infty)$ Range: $(-\infty, +\infty)$

Note the polynomial degree. A cubic ($3x^3 - 5$) has range $(-\infty, +\infty)$ because odd-degree polynomials cover all real outputs. The board briefly wrote “ $[-5, +\infty]$ ” — that would be the range of a *quadratic* like $3x^2 - 5$, whose minimum is -5 . Match the range to the actual degree.

Interval-notation reminder: always write $(-\infty$ and $+\infty)$ with **round** brackets — infinity is never “reached,” so it is never enclosed by a square bracket.

Class 08

Date: June 20

Topic: Rational Equations & Graphing

Rational Equation

Example: $278x + \frac{2}{17.931} = 54.06$

$\Rightarrow x \approx 19.41$

Solving by Expansion

Example: $(2y + 1)(y - 1) = (y + 5)(2y - 5)$

$$2y^2 - 2y + y - 1 = 2y^2 - 5y + 10y - 25$$

$$2y^2 - y - 1 = 2y^2 + 5y - 25$$

$$-y - 1 = 5y - 25$$

$$-1 + 25 = 5y + y$$

$$24 = 6y$$

$$y = 4$$

$$y = 4$$

Cancel the $2y^2$ from both sides first — it appears on the left and right, so it drops out, leaving a linear equation. The board’s line “ $5y = y - 1$ ” skipped collecting terms; carefully moving every term gives $6y = 24$, so **$y = 4$** .

Graphing Basic Equations

Build a table of values, plot the points, then join them. Common shapes:

$y = x$	straight line through origin
$y = x^2$	upward parabola
$y = x^3$	cubic curve (S-shape)
$x = y^2$	sideways parabola (opens right)
$y = x $	V-shape
$y = \frac{1}{x^2}$	two branches, both above the x -axis

Table for $y = |x|$:

x	1	2	3	0	-1	-2	-3	-4	-5
y	1	2	3	0	1	2	3	4	5

(The graph is a V with its vertex at the origin.)

Table for $y = \frac{1}{x^2}$:

x	-4	4	-5	5	6	-6
y	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{25}$	$\frac{1}{25}$	$\frac{1}{36}$	$\frac{1}{36}$

(Always positive, symmetric about the y -axis, approaching 0 as $|x|$ grows.)

Linear Equations & Finding a Point

Example: solve the system $y = 5x + 6$ and $y = 2x$.

$$2x = 5x + 6$$

$$0 = 3x + 6$$

$$3x = -6$$

$$x = -2$$

Then $y = 2x = 2(-2) = -4$.

Intersection: $(x, y) = (-2, -4)$

Careful with the arithmetic: from $0 = 3x + 6$ we get $3x = -6$, so $x = -2$ (not -2 from “ $3x = -2$ ”). Substituting back, $y = 2(-2) = -4$.

Class 09

Date: June 25

Topic: Circles

Standard & General Form

Standard form: $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the centre and r is the radius.

General form: $x^2 + y^2 + 2hx + 2ky + c = 0$, with

centre $(-h, -k)$, radius $r = \sqrt{h^2 + k^2 - c}$.

Example — read the circle

Given: $(x + 3)^2 + (y - 2)^2 = 16$

$$\Rightarrow (x + 3)^2 + (y - 2)^2 = 4^2 \quad (*)$$

Comparing (*) with $(x - h)^2 + (y - k)^2 = r^2$:

$$h = -3, \quad k = 2, \quad r = 4. \quad \text{Centre } (-3, 2), \text{ radius } 4.$$

Example — general to standard form

Given: $x^2 + y^2 + 4x - 6y + 12 = 0$ (1)

$$\begin{aligned} (x^2 + 4x) + (y^2 - 6y) + 12 &= 0 \\ (x^2 + 2 \cdot 2 \cdot x + 2^2) + (y^2 - 2 \cdot 3 \cdot y + 3^2) + 12 - 2^2 - 3^2 &= 0 \\ (x + 2)^2 + (y - 3)^2 + 12 - 4 - 9 &= 0 \\ (x + 2)^2 + (y - 3)^2 &= 1 \end{aligned}$$

$$\text{Centre } (-2, 3), \quad \text{radius } r = \sqrt{1} = 1.$$

Method — completing the square: for each variable, take half the coefficient of the linear term, square it, add it inside and subtract it outside to keep the equation balanced.

Example — intersection points of a circle with the axes

Given: $(x + 3)^2 + (y - 2)^2 = 16$

On the y -axis ($x = 0$):

$$\begin{aligned} (0 + 3)^2 + (y - 2)^2 &= 16 \\ 9 + y^2 - 4y + 4 &= 16 \\ y^2 - 4y + 13 &= 16 \\ y^2 - 4y - 3 &= 0 \end{aligned}$$

On the x -axis ($y = 0$):

$$\begin{aligned} (x + 3)^2 + (0 - 2)^2 &= 16 \\ x^2 + 6x + 9 + 4 &= 16 \\ x^2 + 6x - 3 &= 0 \end{aligned}$$

Each resulting quadratic is then solved (by the quadratic formula) to locate where the circle crosses that axis. This leads directly into the quadratic-equation methods covered in Part 2.

End of Part 1 — continues in Part 2 (11 July onward)

Part 2

Advanced Topics (11 July onward)

July 11 – August 20

What this part covers

Quadratic Equations (4 methods) • Discriminant • Geometry & Pythagoras • Applied Quadratics (screens, pool border, baseball) • Coordinate & Polar Systems • Distance, Midpoint & Slope • Triangle by Coordinates • Functions, Domain & Range • Vertical-Line Test • Even & Odd Functions • Increasing/Decreasing & Extrema • Average Rate of Change & Secant Lines • Intro to Derivatives • Linear Algebra • Gauss-Jordan & Cramer's Rule • Determinants • Exponential Growth Model • Lines, Parabolas & Graph Analysis

Class 10

Date: July 11

Topic: Quadratic Equations

Four Methods to Solve a Quadratic

1. Factoring method
2. Square-root method
3. Completing the square
4. Quadratic formula — the “magic formula,” works in *all* cases:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example — completing the square / square-root method

Solve $x^2 + 6x = 0$:

$$\begin{aligned} x^2 + 2 \cdot x \cdot 3 + 3^2 &= 9 \\ (x + 3)^2 &= 9 \\ x + 3 &= \pm 3 \\ x &= -3 \pm 3 \end{aligned}$$

$$x = 0 \quad \text{or} \quad x = 6$$

Example — quadratic formula

Solve $x^2 + 6x + 0 = 0$ ($a = 1$, $b = 6$, $c = 0$):

$$x = \frac{-6 \pm \sqrt{6^2 - 4(1)(0)}}{2(1)} = \frac{-6 \pm \sqrt{36 - 0}}{2} = \frac{-6 \pm 6}{2}$$

$$x = \frac{0}{2} = 0 \quad (\text{discard? no — valid}) \quad \text{or} \quad x = \frac{-12}{2} = -6. \quad \text{Both roots: } \{0, -6\}.$$

Sign check on the roots. From $\frac{-6 \pm 6}{2}$: the “+” branch gives $\frac{0}{2} = 0$ and the “−” branch gives $\frac{-12}{2} = -6$. Neither root is discarded — both satisfy $x^2 + 6x = 0$. Compare with the completing-the-square version above, which also gives $\{0, -6\}$.

Example — complex roots (negative discriminant)

Solve $3x^2 + 2 = 4x \Rightarrow 3x^2 - 4x + 2 = 0$ ($a = 3$, $b = -4$, $c = 2$):

$$x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(2)}}{2(3)} = \frac{4 \pm \sqrt{16 - 24}}{6} = \frac{4 \pm \sqrt{-8}}{6}$$

Since $\sqrt{-8} = \sqrt{4} \cdot \sqrt{-2} = 2\sqrt{2}i$:

$$x = \frac{4 \pm 2\sqrt{2}i}{6} = \frac{2(2 \pm \sqrt{2}i)}{6} = \frac{2 \pm \sqrt{2}i}{3}$$

$$x = \frac{2 \pm \sqrt{2}i}{3} \quad (\text{two complex conjugate roots})$$

Discriminant $b^2 - 4ac$ tells the nature of the roots: > 0 two real, $= 0$ one repeated, < 0 two complex. Here $16 - 24 = -8 < 0$, so the roots are complex.

Compute the full discriminant $b^2 - 4ac$, not just $-4ac$. Here $(-4)^2 - 4(3)(2) = 16 - 24 = -8$. Forgetting the b^2 term is a common slip.

Class 11

Date: July 16

Topic: Triangles & Pythagorean Applications

Triangle Facts

1. Angle sum: $A + B + C = 180^\circ$
2. Minimum data to define a triangle: 3 sides, *or* 2 sides + 1 angle, *or* 2 angles + 1 side
3. Larger side faces the larger angle: if $A > B$ then side $a >$ side b , etc.

Example — Pythagorean theorem with a variable

Given: hypotenuse $= 4x + 5$; legs $= 3x + 13$ and x .

$$\begin{aligned} a^2 + b^2 &= c^2 \\ x^2 + (3x + 13)^2 &= (4x + 5)^2 \\ x^2 + 9x^2 + 78x + 169 &= 16x^2 + 40x + 25 \\ x^2 + 9x^2 - 16x^2 + 78x - 40x + 169 - 25 &= 0 \\ -6x^2 + 38x + 144 &= 0 \\ 3x^2 - 19x - 72 &= 0 \quad (\div -2) \end{aligned}$$

Factor by splitting the middle term ($3 \cdot (-72) = -216$; -27 and $+8$):

$$\begin{aligned} 3x^2 - 27x + 8x - 72 &= 0 \\ 3x(x - 9) + 8(x - 9) &= 0 \\ (3x + 8)(x - 9) &= 0 \end{aligned}$$

$x = 9$ (taking the positive root; $x = -\frac{8}{3}$ is rejected as a length).
Then: hypotenuse $= 4(9) + 5 = 41$, leg $= 3(9) + 13 = 40$, leg $= 9$. Sides $(9, 40, 41)$.

Square every factor, and take the positive root for a length. $(3x + 13)^2$ expands to $9x^2 + 78x + 169$ (not $3x^2 + \dots$). A side length cannot be negative, so $x = -\frac{8}{3}$ is discarded — keep $x = 9$.

Class 12

Date: July 18

Topic: Applied Quadratics

TV / LCD screen problem

Traditional 4:3 format, diagonal 37": with $L = \frac{4}{3} \cdot (\text{height})$, solve

$$L^2 + \left(\frac{3}{4}L\right)^2 = 37^2 \Rightarrow \frac{25}{16}L^2 = 1369 \Rightarrow L \approx 29.6''$$

$$W = \frac{3}{4}(29.6) \approx 22.2'' \quad \text{Area} = 29.6 \times 22.2 \approx \mathbf{657.12 \text{ sq in}}$$

Widescreen 16:9 format, diagonal 37": with $W = \frac{9}{16}L$,

$$L^2 + \left(\frac{9}{16}L\right)^2 = 37^2 \Rightarrow \frac{337}{256}L^2 = 1369 \Rightarrow L \approx 32.253''$$

$$W = \frac{9}{16}(32.253) \approx 18.142'' \quad \text{Area} = 32.253 \times 18.142 \approx \mathbf{584.97 \text{ sq in}}$$

Takeaway: for the same diagonal, the 4:3 screen has a larger area than the 16:9 widescreen.

Pool border problem

Given: pool diameter 10 ft (so inner radius = 5 ft), border depth 0.25 ft, border volume = 27 cu ft. Let border width = w ; outer radius = $5 + w$.

$$\begin{aligned} V &= \pi (r_{\text{outer}}^2 - r_{\text{inner}}^2) \times h \\ 27 &= \pi [(5 + w)^2 - 5^2] \times 0.25 \\ \frac{108}{\pi} &= (5 + w)^2 - 25 \\ \frac{108}{\pi} + 25 &= (5 + w)^2 \\ 34.377 + 25 &= (5 + w)^2 \\ 59.377 &= (5 + w)^2 \\ 5 + w &= \sqrt{59.377} = 7.706 \\ w &= 2.706 \text{ ft} \end{aligned}$$

Border width $w \approx 2.71$ ft

Baseball diamond problem

Given: the diamond is a square with side 90 ft. Distance from home plate to second base is the square's diagonal.

$$\begin{aligned} d^2 &= 90^2 + 90^2 = 8100 + 8100 = 16200 \\ d &= \sqrt{16200} = \sqrt{90^2 \times 2} = 90\sqrt{2} \approx 90(1.414) = 127.3 \end{aligned}$$

Distance ≈ 127.3 ft

Class 13

Date: mid-July

Topic: Coordinate Systems & Polar Coordinates

Four Coordinate Systems

- | | | |
|-----------------------|---------------------|------|
| 1. Cartesian | (x, y) | } 2D |
| 2. Polar | (r, θ) | |
| 3. Cylindrical | (r, θ, z) | } 3D |
| 4. Spherical | (r, θ, ϕ) | |

Simple graphs: $x = 4$ (vertical line), $y = 2$ (horizontal line), $x = 0$ (y -axis).

Cartesian $\leftrightarrow$ Polar Conversion

$$\begin{aligned} x &= r \cos \theta, & y &= r \sin \theta \\ r &= \sqrt{x^2 + y^2}, & \theta &= \tan^{-1}\left(\frac{y}{x}\right) \end{aligned}$$

Example — point $(4, 2)$ to polar:

$$\begin{aligned} r &= \sqrt{4^2 + 2^2} = \sqrt{16 + 4} = \sqrt{20} \\ \theta &= \tan^{-1}\left(\frac{2}{4}\right) = \tan^{-1}\left(\frac{1}{2}\right) \approx 26.57^\circ \end{aligned}$$

$$(4, 2) \rightarrow (r, \theta) = (\sqrt{20}, 26.57^\circ). \quad \text{Quadrant: I or IV depending on signs.}$$

Example — Cartesian equation to polar

Transform $4xy = 9$ to polar form. Using $x = r \cos \theta$, $y = r \sin \theta$:

$$\begin{aligned} 4(r \cos \theta)(r \sin \theta) &= 9 \\ 4r^2 \sin \theta \cos \theta &= 9 \\ 2r^2(2 \sin \theta \cos \theta) &= 9 \\ 2r^2 \sin 2\theta &= 9 \end{aligned}$$

$$2r^2 \sin 2\theta = 9 \quad (\text{used the identity } 2 \sin \theta \cos \theta = \sin 2\theta)$$

Example — polar equation to Cartesian

Transform $r = 6 \cos \theta$ to Cartesian form. Multiply both sides by r :

$$\begin{aligned} r^2 &= 6r \cos \theta \\ x^2 + y^2 &= 6x \\ x^2 - 6x + y^2 &= 0 \\ (x^2 - 6x + 9) + y^2 &= 9 \\ (x - 3)^2 + y^2 &= 9 \end{aligned}$$

A circle: centre $(3, 0)$, radius 3.

Class 14

Date: mid-July

Topic: Distance, Midpoint, Slope & Triangles

Distance & Midpoint Formulas

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Both differences are squared. The distance formula is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ — the board once wrote $(y_2 - y_1)$ without the square on the first term. Square both.

Example: distance between $(1, 3)$ and $(5, 6)$:

$$d = \sqrt{(5 - 1)^2 + (6 - 3)^2} = \sqrt{16 + 9} = \sqrt{25} = \mathbf{5}$$

Triangle by coordinates

Given $A(-2, 1)$, $B(2, 3)$, $C(3, 1)$: (a) plot, (b) side lengths, (c) verify right triangle, (d) area.

(b) Side lengths:

$$AB = \sqrt{(2 - (-2))^2 + (3 - 1)^2} = \sqrt{4^2 + 2^2} = \sqrt{20} = 2\sqrt{5}$$

$$BC = \sqrt{(3 - 2)^2 + (1 - 3)^2} = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

$$AC = \sqrt{(3 - (-2))^2 + (1 - 1)^2} = \sqrt{5^2 + 0^2} = 5$$

(c) Right-triangle test (Pythagoras): check if $AC^2 = AB^2 + BC^2$:

$$AC^2 = 5^2 = 25, \quad AB^2 + BC^2 = (2\sqrt{5})^2 + (\sqrt{5})^2 = 20 + 5 = 25 \checkmark$$

$AB^2 + BC^2 = AC^2$, so $\triangle ABC$ is a **right triangle** (right angle at B).

(d) Area $= \frac{1}{2} \times BA \times BC = \frac{1}{2} \times 2\sqrt{5} \times \sqrt{5} = \frac{1}{2} \times 2 \times 5 = 5$.

Area = 5 sq units.

Area by determinant

For vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$:

$$\text{Area} = \frac{1}{2} \left| \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \right|$$

For the triangle above:

$$\text{Area} = \frac{1}{2} \left| \det \begin{pmatrix} -2 & 1 & 1 \\ 2 & 3 & 1 \\ 3 & 1 & 1 \end{pmatrix} \right| = \frac{1}{2} \times 10 = \mathbf{5 \text{ sq units}}$$

Slope & Perpendicularity

Slope $m = \frac{y_2 - y_1}{x_2 - x_1}$. Two lines are perpendicular if $m_1 m_2 = -1$; parallel if $m_1 = m_2$.

For $\triangle ABC$ above: $m_{AB} = \frac{3 - 1}{2 - (-2)} = \frac{2}{4} = \frac{1}{2}$, $m_{BC} = \frac{1 - 3}{3 - 2} = \frac{-2}{1} = -2$.

$$m_{AB} \cdot m_{BC} = \frac{1}{2} \times (-2) = -1$$

Product of slopes = -1 , confirming $\triangle ABC$ is a right triangle.

Class 15

Date: July 25

(Quiz 2 was scheduled 30 July)

Topic: Functions, Domain & Range

Function — Definition

Domain = input of a function. **Range** = output of a function.

A function is a relation in which every input (domain value) is assigned to exactly one output (range value):

1. Every input must have an output.
2. Each input can have only one output.

Example — $y = f(x) = x^3$ over $\{1, 2, 3\}$:

Domain	$\{1, 2, 3\}$
Range	$\{1, 8, 27\}$
Co-domain	$\{1, 8, 27, 64\}$

Determine whether a relation is a function

(a) $\{(1, 4), (2, 5), (3, 6), (4, 7)\}$

Domain $\{1, 2, 3, 4\}$, Range $\{4, 5, 6, 7\}$. Every input has one output $\Rightarrow$ **function**.

(c) $\{(-3, 9), (-2, 4), (0, 0), (1, 1), (-3, 8)\}$

-3 maps to both 9 and $8 \Rightarrow$ **not a function**.

Domain of $\frac{5}{x-2}$

Domain = $\{x \mid x \neq 2\}$

Vertical-Line Test (from a graph)

Vertical-line test: if any vertical line meets the graph more than once, it is *not* a function. $x^2 + y^2 = 1$ (a circle) fails the test — solving gives $y = \pm\sqrt{1-x^2}$, two outputs per x .

Evaluating a piecewise/rational function

Example: $f(x) = 3x^2 - 5x + \frac{1}{x}$ at $x = -2$:

$$f(-2) = 3(-2)^2 - 5(-2) + \frac{1}{-2} = 3(4) + 10 - \frac{1}{2} = 12 + 10 - \frac{1}{2} = 22 - \frac{1}{2} = \frac{43}{2}$$

$$f(-2) = \frac{43}{2} = 21.5$$

Domain of $g(x)$ with factor $x + 4$ in the denominator:

$$x + 4 \neq 0 \Rightarrow x \neq -4; \text{ domain} = \{x \mid x \neq -4\} = (-\infty, -4) \cup (-4, \infty)$$

Class 16

Date: August 01

Topic: Even & Odd Functions

Definitions

Odd: for every x in the domain, $-x$ is also in the domain, and $f(-x) = -f(x)$.

Even: $f(-x) = f(x)$.

Shortcut (symmetry): Even $\Rightarrow$ symmetric about the y -axis. Odd $\Rightarrow$ symmetric about the origin.

Example — classify each function

(a) $f(x) = x^2 - 5$:

$$f(-x) = (-x)^2 - 5 = x^2 - 5 = f(x) \Rightarrow \text{even}$$

(b) $g(x) = x^3 - 1$:

$$g(-x) = (-x)^3 - 1 = -x^3 - 1 = -(x^3 + 1) \neq \pm g(x) \Rightarrow \text{neither}$$

(c) $h(x) = 5x^3 - x$:

$$h(-x) = 5(-x)^3 - (-x) = -5x^3 + x = -(5x^3 - x) = -h(x) \Rightarrow \text{odd}$$

(d) $f(x) = |x|$:

$$f(-x) = |-x| = |x| = f(x) \Rightarrow \text{even}$$

(e) $i(x) = \cos x$:

$$i(-x) = \cos(-x) = \cos x \Rightarrow \text{even}$$

Example — even/odd from a graph

A graph symmetric about the y -axis is **even**; symmetric about the origin is **odd**; a graph with no symmetry is **neither**.

Class 17

Date: August 06

Topic: Slopes, Increasing/Decreasing & Local Extrema

Four Kinds of Slope

1. **Positive** rises left-to-right
2. **Negative** falls left-to-right
3. **Zero** $m = \frac{\Delta y}{\Delta x} = 0$ (horizontal line)
4. **Undefined** $m = \frac{\Delta y}{\Delta x} = \text{undefined}$ (vertical line, $\Delta x = 0$)

Local Maximum & Local Minimum (peaks / hills / valleys)

A **local maximum** is a peak (hill); a **local minimum** is a valley. When reading coordinates for a max/min value, we report the **y-value** (the output at the peak/valley).

Absolute Maximum & Minimum

Example — from a graph passing through $(0, 1)$, $(3, 6)$, $(4, 4)$, $(5, 5)$ on domain $[0, 5]$:

Domain = $[0, 5]$, Range = $[1, 6]$.
 Absolute **maximum** = 6 at $x = 3$. Absolute **minimum** = 1 at $x = 0$.

Reading a graph value: “at $x = 0$, $f(x) = y = 1$ ” means the output (the y -coordinate) is the max/min value; the x -coordinate is *where* it occurs.

Class 18

Date: August 08
Derivatives

Topic: Average Rate of Change, Secant Lines & Intro to

Rate of Change

Slope = rate of change of one variable with respect to another. For $y = f(x)$:

$$\frac{dy}{dx} = \frac{\text{dependent}}{\text{independent}}$$

Point-slope form: $y - y_1 = m(x - x_1)$, slope-intercept form: $y = mx + c$.

Average Rate of Change

Average rate of change = $\frac{f(b) - f(a)}{b - a}$ (the slope of the **secant line** joining the two points).

Example: average rate of change of $g(x) = 3x^2 - 2x + 3$ from $x = -2$ to $x = 1$:

$$\begin{aligned} g(1) &= 3(1)^2 - 2(1) + 3 = 3 - 2 + 3 = 4 \\ g(-2) &= 3(-2)^2 - 2(-2) + 3 = 12 + 4 + 3 = 19 \\ \text{Avg rate} &= \frac{g(1) - g(-2)}{1 - (-2)} = \frac{4 - 19}{3} = \frac{-15}{3} = -5 \end{aligned}$$

Average rate of change = -5

Secant Line Equation

Example: the secant line through $(-2, g(-2)) = (-2, 19)$ and $(1, g(1)) = (1, 4)$ has slope $m_{\text{sec}} = -5$. Point-slope form:

$$\begin{aligned}y - y_1 &= m_{\text{sec}}(x - x_1) \\y - 19 &= -5(x - (-2)) \\y - 19 &= -5x - 10 \\y &= -5x + 9\end{aligned}$$

Secant line: $y = -5x + 9$

Intro to the Derivative

The **derivative** gives the instantaneous rate of change (slope of the tangent). Power rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Example — critical points of $g(x) = x^3 - 27x$:

$$\begin{aligned}g'(x) &= \frac{dg}{dx} = 3x^2 - 27 \\g'(x) = 0 &\Rightarrow 3x^2 - 27 = 0 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3\end{aligned}$$

Critical points at $x = 3$ (local minimum) and $x = -3$ (local maximum).
At $x = -3$: $g(-3) = 54$; at $x = 3$: $g(3) = -54$.

Solve $x^2 = 9$ with $\pm$. Taking the square root gives $x = \pm 3$, *both* roots — stopping at $x = 3$ misses the second critical point. Since g' changes sign, one point is a max and the other a min.

Even/odd check and x -intercepts of $f(x) = -x^4 + 8x^2 + 8$

Even test:

$$f(-x) = -(-x)^4 + 8(-x)^2 + 8 = -x^4 + 8x^2 + 8 = f(x) \Rightarrow \text{even}$$

y -intercept ($x = 0$): $f(0) = 8$. **x -intercepts:** solve $-x^4 + 8x^2 + 8 = 0$:

$x \approx \pm 2.9831$ (and the inner turning-point structure gives ± 0.748 as the other pair from the related factorization). The curve is symmetric about the y -axis (even).

Class 19

Date: August 13

Topic: Linear Algebra

Linear vs. Non-linear

An expression/equation is **linear** if the power of every variable is exactly 1 (i.e. degree 0 in the sense of no higher powers, no roots, no products of variables, no trig, no log). **If the power is 1, it is linear.**

Linear	Non-linear
$4x + 5 = 0$	$4x^2 + 5 = 0$
$4x^1 + 5x^0 = 0$	$4xy + 5 = 0$
	$4\sqrt{x} + 5 = 0$ (i.e. $4x^{1/2} + 5 = 0$)
	$4\sin x + 5 = 0, \quad 4\log x + 5 = 0$
	$\frac{dy^2}{dx} + 5 = 0, \quad 5 + \left(\frac{dy}{dx}\right)\left(\frac{dy}{dx}\right) = 0$

Note: $4x + 5$ is an algebraic *expression*; $4x + 5 = 0$ is an *equation*.

Matrix Forms

- **Row Echelon Form (REF)** — via **Gauss elimination**; leading 1s with zeros below.
- **Reduced Row Echelon Form (RREF)** — via **Gauss-Jordan elimination**; leading 1s with zeros both above and below.

Identity matrix (an RREF):

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{echelon example: } \begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 7 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

System of Linear Equations & the Augmented Matrix

System: $7x_1 - 2x_2 = 3, \quad 3x_1 + x_2 = 5.$

Matrix form: $Ax = B$, where A is the **coefficient matrix**, x the variable vector, B the constants.

$$A = \begin{pmatrix} 7 & -2 \\ 3 & 1 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad B = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

The **augmented matrix**:

$$[A \mid B] = \left(\begin{array}{cc|c} 7 & -2 & 3 \\ 3 & 1 & 5 \end{array} \right)$$

Solving by Gauss-Jordan Elimination

Step 1 — $R_1 \rightarrow \frac{R_1}{7}$ (create leading 1):

$$\left(\begin{array}{cc|c} 1 & -\frac{2}{7} & \frac{3}{7} \\ 3 & 1 & 5 \end{array} \right)$$

Step 2 — $R_2 \rightarrow R_2 - 3R_1$ (zero below the leading 1):

$$R_2 : \quad (3 - 3(1), 1 - 3(-\frac{2}{7}), 5 - 3(\frac{3}{7})) = (0, \frac{13}{7}, \frac{26}{7})$$

$$\left(\begin{array}{cc|c} 1 & -\frac{2}{7} & \frac{3}{7} \\ 0 & \frac{13}{7} & \frac{26}{7} \end{array} \right)$$

Step 3 — $R_2 \rightarrow R_2 \times \frac{7}{13}$ (leading 1 in row 2):

$$\left(\begin{array}{cc|c} 1 & -\frac{2}{7} & \frac{3}{7} \\ 0 & 1 & 2 \end{array} \right)$$

This gives the equations:

$$x_1 - \frac{2}{7}x_2 = \frac{3}{7}, \quad x_2 = 2$$

Back-substitute $x_2 = 2$ into the first equation:

$$x_1 = \frac{3}{7} + \frac{2}{7}(2) = \frac{3}{7} + \frac{4}{7} = \frac{7}{7} = 1$$

$$(x_1, x_2) = (1, 2)$$

Solving the Same System by Cramer's Rule

$$x_i = \frac{\det(A_i)}{\det(A)}, \quad \text{where } A_i \text{ is } A \text{ with column } i \text{ replaced by } B.$$

Coefficient matrix and replacements:

$$A = \begin{pmatrix} 7 & -2 \\ 3 & 1 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 3 & -2 \\ 5 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 7 & 3 \\ 3 & 5 \end{pmatrix}$$

$$\det(A) = (7)(1) - (-2)(3) = 7 + 6 = 13$$

$$\det(A_1) = (3)(1) - (-2)(5) = 3 + 10 = 13$$

$$\det(A_2) = (7)(5) - (3)(3) = 35 - 9 = 26$$

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{13}{13} = 1, \quad x_2 = \frac{\det(A_2)}{\det(A)} = \frac{26}{13} = 2$$

$$(x_1, x_2) = (1, 2) \quad \text{— matches the Gauss-Jordan result.}$$

Instructor's convention: determinants are computed by strict **first-row cofactor expansion** (not "smart" row/column selection). For 3×3 and larger matrices, always expand along the first row.

Determinant by First-Row Expansion (larger system)

Example — system $4x + 5y = 2$, $11x + y = 3$, $x + 5y + 2z = 1$, coefficient matrix:

$$A = \begin{pmatrix} 4 & 5 & 0 \\ 11 & 1 & 2 \\ 1 & 5 & 2 \end{pmatrix}$$

Expanding along the first row:

$$\begin{aligned} \det(A) &= 4 \det \begin{pmatrix} 1 & 2 \\ 5 & 2 \end{pmatrix} - 5 \det \begin{pmatrix} 11 & 2 \\ 1 & 2 \end{pmatrix} + 0 \det \begin{pmatrix} 11 & 1 \\ 1 & 5 \end{pmatrix} \\ &= 4(2 - 10) - 5(22 - 2) + 0 \\ &= 4(-8) - 5(20) \\ &= -32 - 100 = -132 \end{aligned}$$

$$\det(A) = -132$$

Larger Determinants (for exam)

For a 4×4 matrix such as

$$A = \begin{pmatrix} -1 & -4 & 2 & 1 \\ 2 & -1 & 7 & 9 \\ -1 & 1 & 3 & 1 \\ 1 & -2 & 1 & -4 \end{pmatrix},$$

the same first-row cofactor expansion is applied recursively: expand along row 1 into four 3×3 determinants, each of which is then expanded along *its* first row.

Class 20

Date: August 15

Topic: Exponential Growth Model

Building the Model

Let t = time, N_0 = initial population, k = proportional constant (**growth rate**):

$$k = \text{birth rate} - \text{death rate} + \text{migration} = b - d + m$$

If the rate of change of the population is proportional to the current population:

$$\begin{aligned} \frac{dN}{dt} &\propto N \\ \frac{dN}{dt} &= kN \\ \frac{dN}{N} &= k dt \\ \int \frac{dN}{N} &= \int k dt \\ \ln N &= kt + C \\ N &= e^{kt+C} \\ N(t) &= e^{kt} \cdot e^C = N_0 e^{kt} \end{aligned}$$

Malthusian growth model: $N(t) = N_0 e^{kt}$

Example — finding the growth rate k

Given: initial population $N_0 = 1000$; after $t = 1$ day, $N(1) = 1800$.

$$\begin{aligned} N(1) &= N_0 e^{k(1)} \\ 1800 &= 1000 e^k \\ e^k &= \frac{1800}{1000} = 1.8 \\ k &= \ln(1.8) \approx 0.5878 \end{aligned}$$

Population after $t = 3$ days:

$$N(3) = N_0 e^{k \cdot 3} = 1000 e^{0.5878 \times 3} \approx \mathbf{5832}$$

$$k \approx 0.5878; \quad N(3) \approx 5832$$

Example — finding the time to reach a target

Given: target $N(t) = 10,000$; growth rate $k = 0.5878$ (from above), $N_0 = 1000$.

$$\begin{aligned} N(t) &= N_0 e^{kt} \\ e^{kt} &= \frac{N(t)}{N_0} = \frac{10,000}{1000} = 10 \\ kt &= \ln(10) \\ t &= \frac{\ln(10)}{0.5878} = \frac{2.3026}{0.5878} \approx 3.917 \approx 3.92 \text{ days} \approx 4 \text{ days} \end{aligned}$$

$$t \approx 3.92 \text{ days} \approx 4 \text{ days}$$

Finding k when the data point is not at $t = 1$: if the given value were at, say, $t = 3$, you must divide the exponent by 3 — i.e. $e^{3k} = \text{ratio}$, so $k = \frac{1}{3} \ln(\text{ratio})$. Here the data point is at $t = 1$, so $k = \ln(1.8)$ directly.

Class 21

Date: August 20

Topic: Lines, Parabolas & Graph Analysis

Problem A — reading a line and a parabola from a graph

Setup: the graph shows a straight line $y = mx + c$ crossing a parabola $y = ax^2 + b$. The line passes through $(0, -2)$ and $(2, 0)$; the parabola has x -intercepts $(2, 0)$ and $(5, 0)$.

(1) Find the equation of the straight line. Through $(0, -2)$ and $(2, 0)$:

$$m = \frac{0 - (-2)}{2 - 0} = \frac{2}{2} = 1$$

With y -intercept $c = -2$:

$$\text{Line: } y = x - 2$$

(2) Find the equation of the parabola. With x -intercepts at $x = 2$ and $x = 5$, the factored form is $y = a(x - 2)(x - 5)$. Taking $a = 1$:

$$y = (x - 2)(x - 5) = x^2 - 7x + 10$$

$$\text{Parabola: } y = x^2 - 7x + 10$$

(3) Find the other intersection point of the line and the parabola. Set the line equal to the parabola:

$$\begin{aligned}x - 2 &= x^2 - 7x + 10 \\0 &= x^2 - 8x + 12 \\0 &= (x - 2)(x - 6)\end{aligned}$$

So $x = 2$ or $x = 6$. Since $x = 2$ is already known (it sits on the x -axis where both meet), the *other* intersection uses $x = 6$:

$$y = x - 2 = 6 - 2 = 4$$

Other intersection point: (6, 4)

Don't stop at the first root. A line meets a parabola in up to two points; after finding $x = 2$, solve for the second root ($x = 6$) and substitute back to get its y -value. Missing the second intersection is a common slip on graph-reading questions.

Problem A (continued) — even / odd check

Determine whether $h(x) = x^2 + 7x + 5$ is even, odd, or neither.

$$h(-x) = (-x)^2 + 7(-x) + 5 = x^2 - 7x + 5$$

Compare: $h(-x) = x^2 - 7x + 5 \neq h(x)$ (so not even); and $-h(x) = -(x^2 + 7x + 5) = -x^2 - 7x - 5 \neq h(-x)$ (so not odd).

$h(x) = x^2 + 7x + 5$ is **neither even nor odd**.

Why: a quadratic is even only when it has no odd-power term. The $+7x$ term breaks y -axis symmetry, and the constant $+5$ breaks origin symmetry, so it is neither.

Problem B — full analysis of $f(x) = x^2 - 6x + 5$

(1) Find the x - and y -intercepts.

x -intercepts — set $f(x) = 0$:

$$\begin{aligned}x^2 - 6x + 5 &= 0 \\(x - 1)(x - 5) &= 0\end{aligned}$$

So $x = 1$ or $x = 5$.

x -intercepts: (1, 0) and (5, 0)

y -intercept — put $x = 0$:

$$f(0) = 0^2 - 6(0) + 5 = 5$$

y -intercept: (0, 5)

(2) Find the vertex. Using $x = -\frac{b}{2a}$ with $a = 1$, $b = -6$:

$$x = \frac{-(-6)}{2(1)} = \frac{6}{2} = 3$$

$$f(3) = 3^2 - 6(3) + 5 = 9 - 18 + 5 = -4$$

Vertex: $(3, -4)$

(3) State the domain and range.

Domain = $(-\infty, \infty)$; Range = $[-4, \infty)$ (opens upward, minimum at $y = -4$).

Interval-notation reminder: $-\infty$ and ∞ always take **round** brackets; the minimum value -4 *is* attained, so it takes a **square** bracket: $[-4, \infty)$.

End of Part 2 — End of the full NSC4182 class-notes compilation.